

Probabilistic Numerics

Probabilistic numerical computation formalises the connection between machine learning and applied mathematics. Numerical algorithms approximate intractable quantities from computable ones. They estimate integrals from evaluations of the integrand, or the path of a dynamical system described by differential equations from evaluations of the vector field. In other words, they infer a latent quantity from data. This book shows that it is thus formally possible to think of computational routines as learning machines, and to use the notion of Bayesian inference to build more flexible, efficient, or customised algorithms for computation.

The text caters for Masters' and PhD students, as well as postgraduate researchers in artificial intelligence, computer science, statistics, and applied mathematics. Extensive background material is provided along with a wealth of figures, worked examples, and exercises (with solutions) to develop intuition.

Philipp Hennig holds the Chair for the Methods of Machine Learning at the University of Tübingen, and an adjunct position at the Max Planck Institute for Intelligent Systems. He has dedicated most of his career to the development of Probabilistic Numerical Methods. Hennig's research has been supported by Emmy Noether, Max Planck and ERC fellowships. He is a co-Director of the Research Program for the Theory, Algorithms and Computations of Learning Machines at the European Laboratory for Learning and Intelligent Systems (ELLIS).

Michael A. Osborne is Professor of Machine Learning at the University of Oxford, and a co-Founder of Mind Foundry Ltd. His research has attracted £10.6M of research funding and has been cited over 15,000 times. He is very, very Bayesian.

Hans P. Kersting is a postdoctoral researcher at INRIA and École Normale Supérieure in Paris, working in machine learning with expertise in Bayesian inference, dynamical systems, and optimisation.

‘This impressive text rethinks numerical problems through the lens of probabilistic inference and decision making. This fresh perspective opens up a new chapter in this field, and suggests new and highly efficient methods. A landmark achievement!’

– **Zoubin Ghahramani, University of Cambridge**

‘In this stunning and comprehensive new book, early developments from Kac and Larkin have been comprehensively built upon, formalised, and extended by including modern-day machine learning, numerical analysis, and the formal Bayesian statistical methodology. Probabilistic numerical methodology is of enormous importance for this age of data-centric science and Hennig, Osborne, and Kersting are to be congratulated in providing us with this definitive volume.’

– **Mark Girolami, University of Cambridge and The Alan Turing Institute**

‘This book presents an in-depth overview of both the past and present of the newly emerging area of probabilistic numerics, where recent advances in probabilistic machine learning are used to develop principled improvements which are both faster and more accurate than classical numerical analysis algorithms. A must-read for every algorithm developer and practitioner in optimization!’

– **Ralf Herbrich, Hasso Plattner Institute**

‘Probabilistic numerics spans from the intellectual fireworks of the dawn of a new field to its practical algorithmic consequences. It is precise but accessible and rich in wide-ranging, principled examples. This convergence of ideas from diverse fields in lucid style is the very fabric of good science.’

– **Carl Edward Rasmussen, University of Cambridge**

‘An important read for anyone who has thought about uncertainty in numerical methods; an essential read for anyone who hasn’t’

– **John Cunningham, Columbia University**

‘This is a rare example of a textbook that essentially founds a new field, re-casting numerics on stronger, more general foundations. A tour de force.’

– **David Duvenaud, University of Toronto**

‘The authors succeed in demonstrating the potential of probabilistic numerics to transform the way we think about computation itself.’

– **Thore Graepel, Senior Vice President, Altos Labs**

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CAMBRIDGE
UNIVERSITY PRESS

University Printing House, Cambridge CB2 8BS, United Kingdom
One Liberty Plaza, 20th Floor, New York, NY 10006, USA
477 Williamstown Road, Port Melbourne, VIC 3207, Australia
314–321, 3rd Floor, Plot 3, Splendor Forum, Jasola District Centre, New Delhi – 110025, India
103 Penang Road, #05-06/07, Visioncrest Commercial, Singapore 238467

Cambridge University Press is part of the University of Cambridge.
It furthers the University's mission by disseminating knowledge in the pursuit of
education, learning, and research at the highest international levels of excellence.

www.cambridge.org
Information on this title: www.cambridge.org/9781107163447
DOI: [10.1017/9781316681411](https://doi.org/10.1017/9781316681411)

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First published 2022

Printed in the United Kingdom by TJ Books Limited, Padstow Cornwall

A catalogue record for this publication is available from the British Library.

ISBN 978-1-107-16344-7 Hardback

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